

Memory Hierarchy 5



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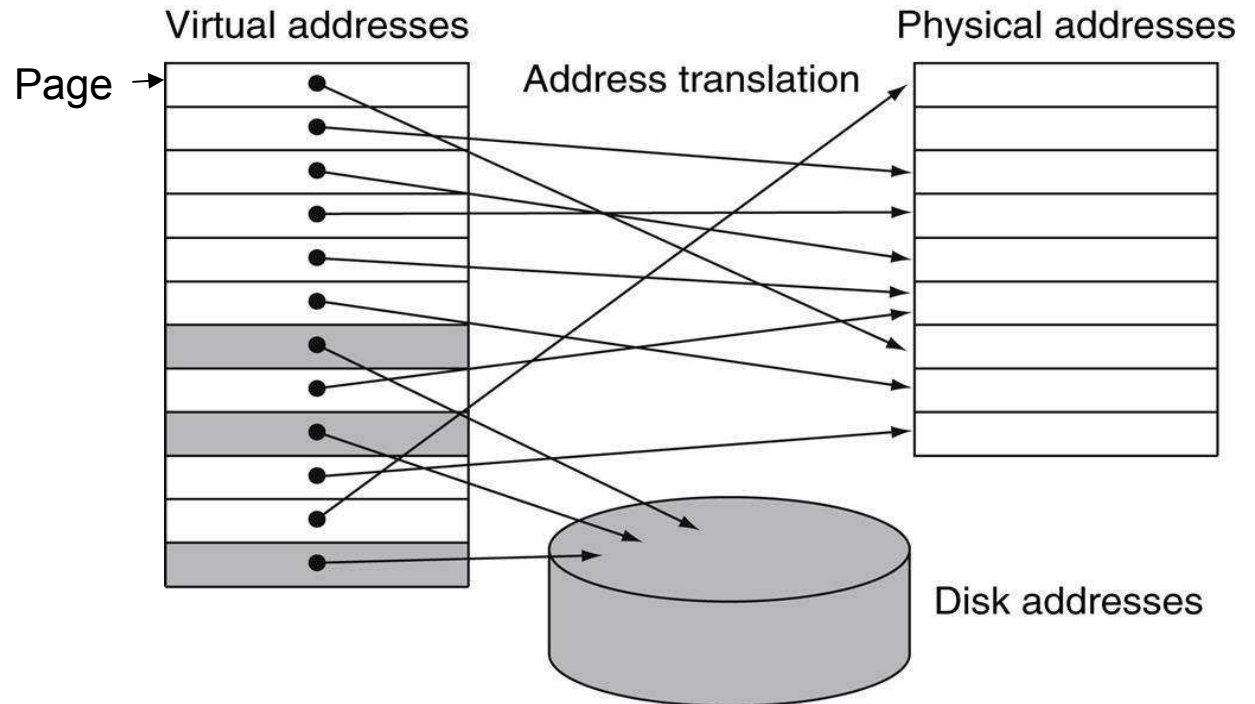
Outline

- Virtual memory recap and examples

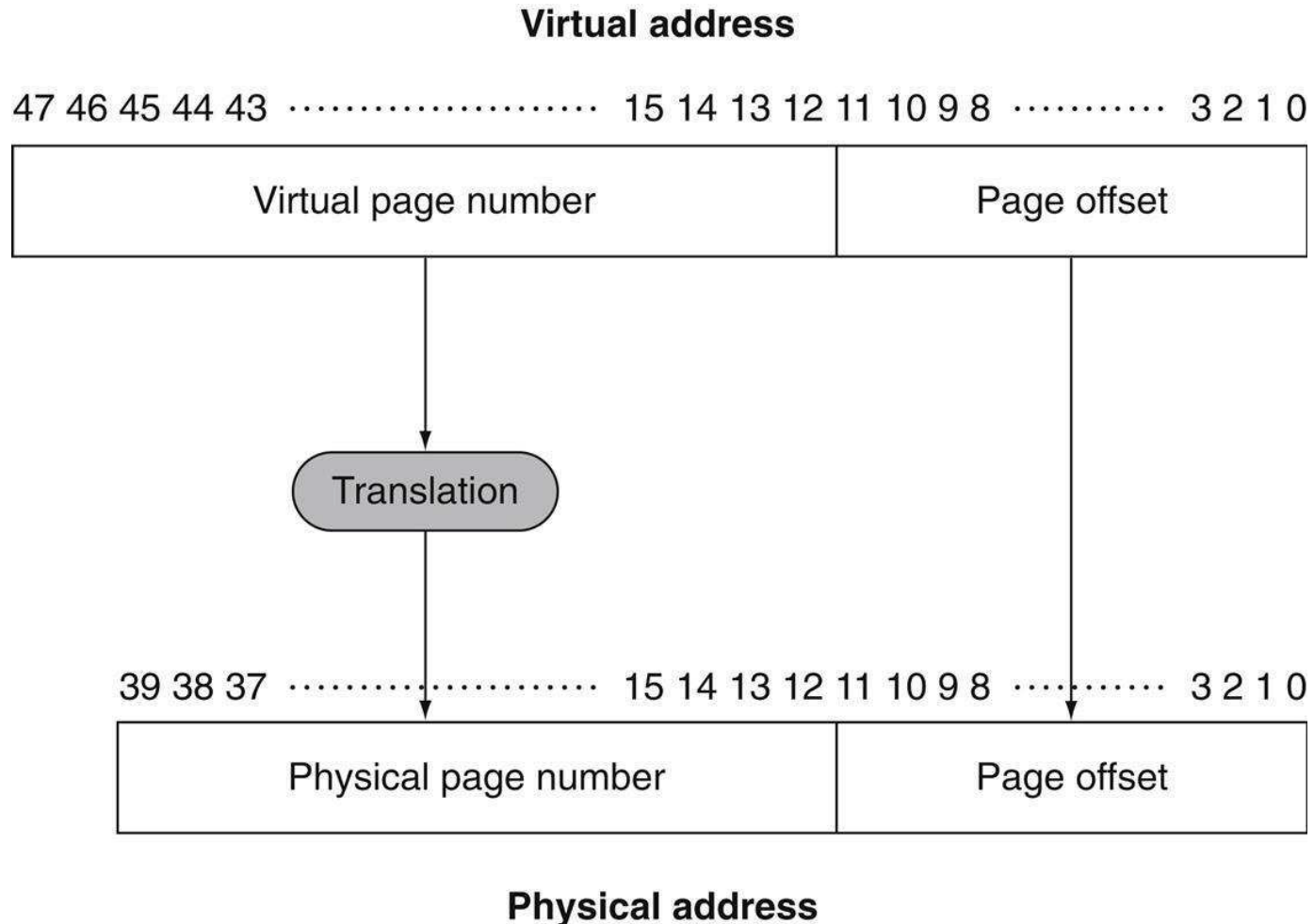


Virtual Memory - recap

- Main memory act as “cache” for secondary storage
- Motivation
- Protection
- Terminology

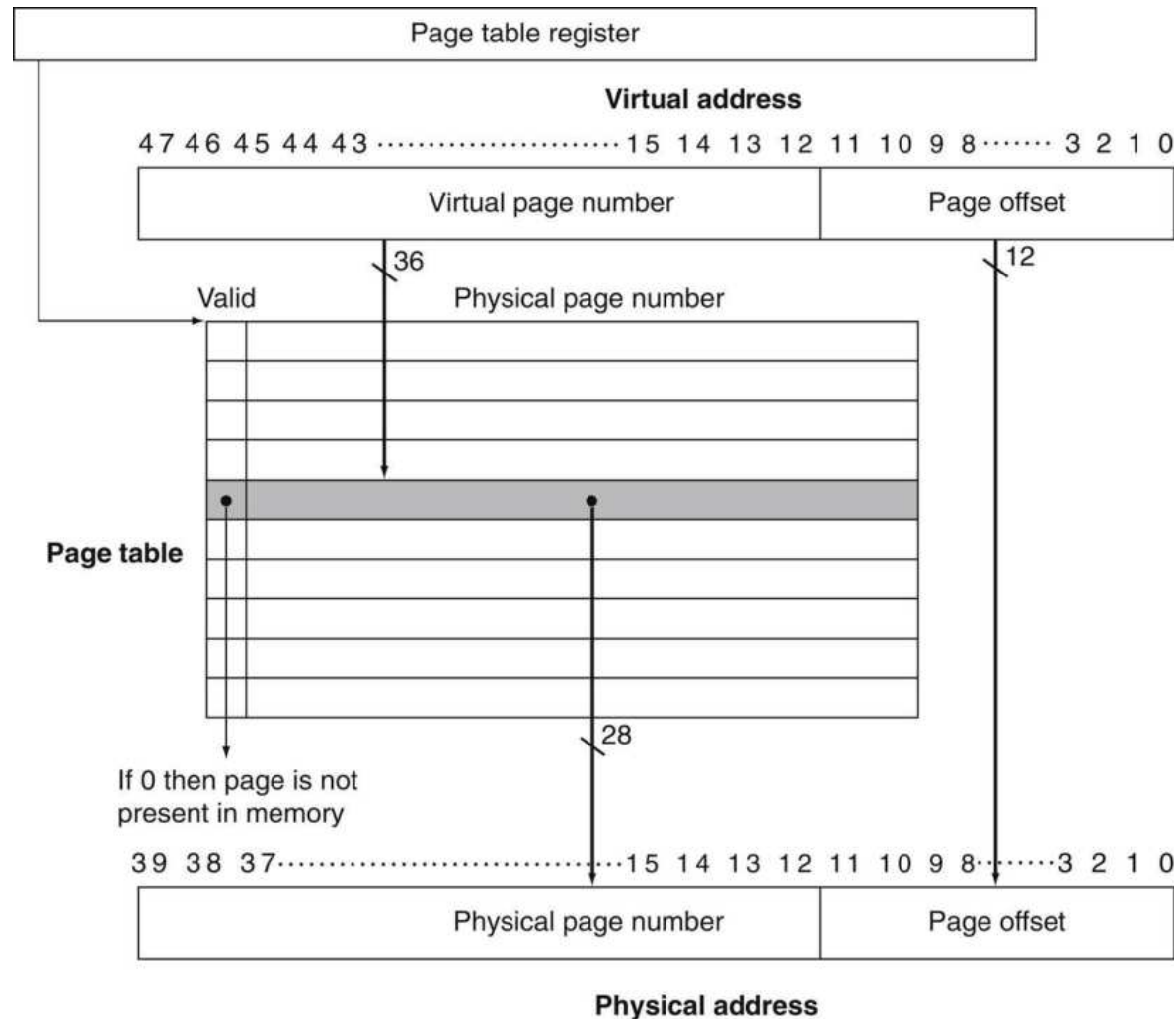


Address translation



Virtual memory configuration

- Locate pages using a table that indexes the main memory = **page table**
- Page table resides in main memory
- Valid bit = 0 → page fault
 - OS should manage
 - Swap space

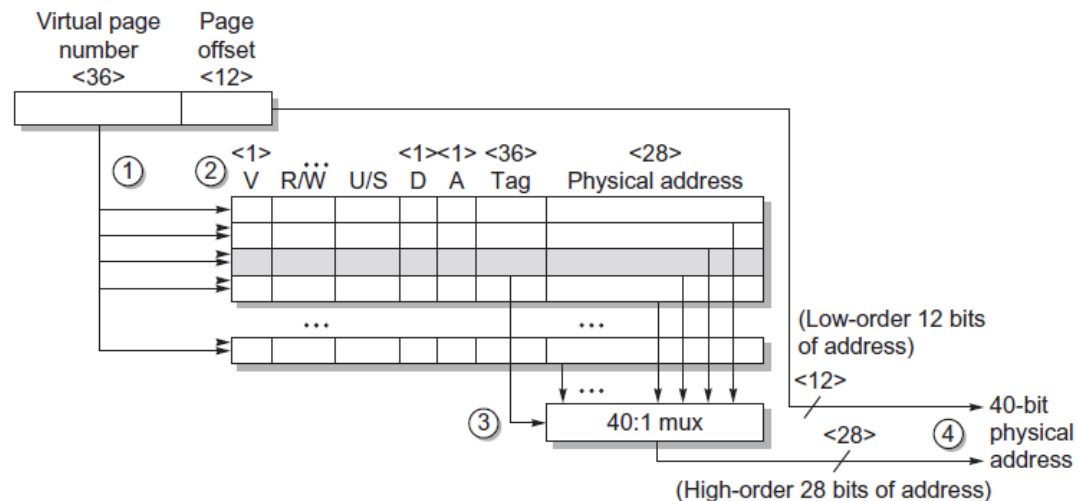


Page Table

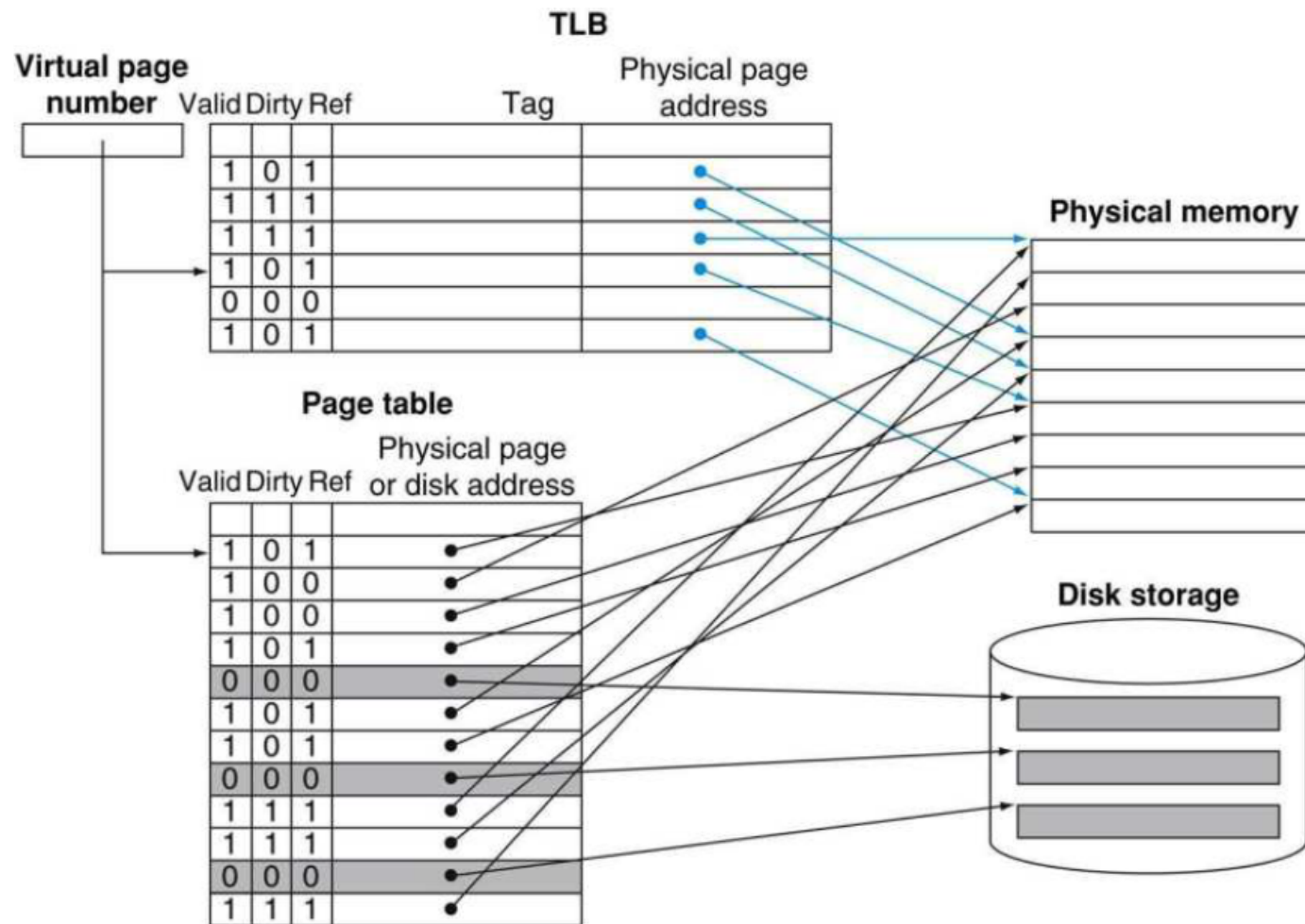
- ❏ Maps virtual address to physical address
- ❏ Stored in main memory
- ❏ Virtual page number to index page table entry
 - Assume Byte-addressing
 - 4KiB pages $\rightarrow 2^{12}$ bytes in a page
 - 48-bit Virtual address $\rightarrow$ 12 bit page offset
 - 36 bits for indexing page table entries
 - 2^{36} page table entries $\sim$ 64 billion entries!
- ❏ Techniques to reduce page table size
 - E.g. page the page table, multi-level table

Fast Address Translation

- Paging → 2 memory accesses for every memory reference by CPU
 - Obtain physical address
 - Access data/instruction
- Translation Look-aside Buffer (TLB)
 - A special address translation cache



Translation Look-aside Buffer

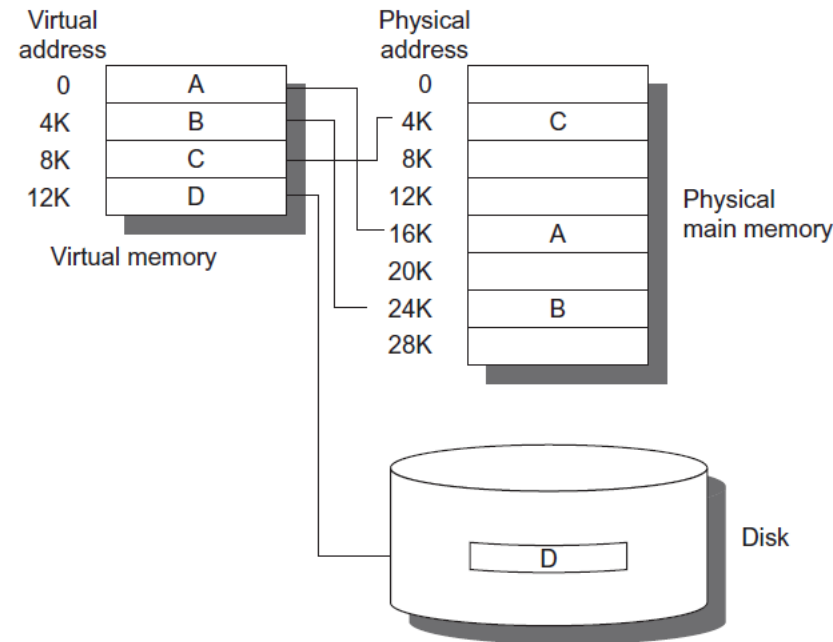


TLB

- A special cache to keep address translations
- Accesses within a page may have
 - Temporal locality
 - Spatial locality
- Reduce memory accesses needed to lookup the page table
- TLB includes additional status bits
 - dirty bit, reference bit

Page fault handling

1. Hardware detects a nonresident page and **traps** to the operating system
2. OS identifies the required virtual page and locates it in secondary storage
3. OS selects a physical page frame, potentially evicting another page
4. Required page is transferred into physical memory
5. Page-table and TLB state are updated
6. The faulting instruction is restarted



Impact of page size

- Page table size – inversely proportional to page size
- Transferring larger pages → more efficient
 - to or from secondary storage, possibly over a network
- Larger pages → more memory mapped in TLB
 - Less TLB misses
- Smaller pages → conserve memory
 - unused memory in large pages → **internal fragmentation**

Protection via virtual memory

- **Process** – running program and its states
 - *A program's breathing air and living space*
- **Process switch/ context switch** – switch from one process to another
 - Time-sharing: several interactive programs share processor and memory
- OS and Computer Architecture to maintain correct process behavior
 - Computer designer to make sure processors can store process state in hardware
 - OS designer to make sure processes do not interfere each other's computation

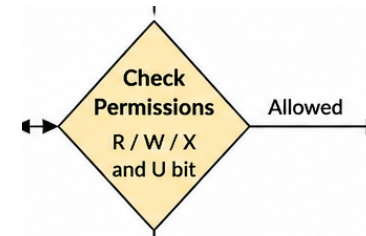
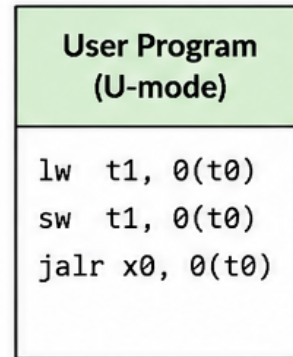
Context switching

- Copy current state to disk?
 - Safe but unacceptably slow
- OS to partition memory and keep state of multiple processes at the same time
 - Computer designer to provide protection from one process modifying state of another process in memory
 - E.g. own page table, privilege and user modes
- Sharing code and data between processes

RISC-V Memory Protection Example

Examples (from U-mode)

1) Read user data VA: 0x40000000 (load word)	PASS R=1, U=1
2) Write user data VA: 0x40000000 (store word)	PASS W=1, U=1
3) Write code page VA: 0x40001000 (store word)	FAULT W=0 (write not allowed)
4) Read kernel page VA: 0xFFFFFFFF80000000 (load word)	FAULT U=0 (user not allowed)
5) Execute data page VA: 0x40000000 (jalr)	FAULT X=0 (execute not allowed)



Page Table (Sv39)

VPN	PPN	R	W	X	U
0x0000000004000000	0x00100000	1	1	0	1
0x0000000004000100	0x00101000	1	0	1	1
0xFFFFFFFF80000000	0x01000000	1	1	1	0
...					

Virtual Machines

- A foundational technology for cloud computing
- Computer architecture support
 - At least two modes of access: User vs Supervisor
 - Only OS can change certain processor states such as mode, exceptions and memory protection bits
 - A mechanism for processor to move between modes
 - A mechanism to limit memory accesses
- Page table based protection
 - includes protection bits
 - Only OS can change page table
- OS is large code → more bugs: **Virtual Machine!**

Protection via Virtual Machines

- First in 1960 main frames recently into personal computers
 - Demand for isolation and security
 - Standard OS's security and reliability failures
 - Sharing single computers among unrelated users (cloud)
 - VM overhead is bearable due to high processor speeds

From virtual memory to virtual machines

- ❑ Virtual memory virtualizes the memory address space of a process.
- ❑ A virtual machine virtualizes an entire computer interface at the ISA/system level.
- ❑ Multiple guest operating systems can share one physical machine.
- ❑ A virtual machine monitor (VMM), also called a hypervisor, controls the shared hardware.
- ❑ The VMM must preserve isolation while allowing efficient execution

What a VMM must provide

- ❑ Isolation: one guest must not access another guest's resources.
- ❑ Correctness: guest software should observe the expected machine behavior.
- ❑ Resource control: the VMM must retain control over processor, memory, and I/O resources.
- ❑ Efficiency: most ordinary guest instructions should execute directly on the hardware.
- ❑ These requirements motivate hardware support for virtualization.

System virtual machines

- A system VM presents a complete machine environment to a guest OS.
- The guest OS behaves as if it owns a machine.
- Several guests can run concurrently on the same physical host.
- Virtual machines can support legacy software, isolation, consolidation, testing, and cloud infrastructure.
- The VMM multiplexes physical resources among guests.

Virtualizable architecture

- An architecture is easier to virtualize when sensitive operations can be controlled by the VMM.
- Ordinary instructions should execute directly whenever safe.
- Operations that could compromise control must trap or otherwise be intercepted.
- ISA design therefore has a major effect on virtualization overhead.
- Modern processor ISAs include explicit hardware mechanisms to make virtualization practical.

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THANK YOU